AI-Assisted Learning and the Reversal of Long-Run Skill Degradation

Final presentation — Track A

Working results for class critique

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https://github.com/wbgradost/ai-project

Can AI improve long-run skill while reducing human effort?

Result in one sentence

With $b, \lambda_0, \kappa, \mu > 0$ and fixed primitives, AI initially raises the stationary high-skill share if and only if $\eta b > 1$, where b is effort without AI and η measures learning complementarity.

The improvement is bounded: assistance eventually eliminates effort, leaving only background learning.

$$\lambda(e, a) = \lambda_0 + \kappa e(1 + \eta a).$$

Baseline and contribution

Aouad, Lykouris, and Zhong (2026), Proposition 3.5: when upward learning rates depend only on effort, more assistance weakly shifts stationary skill downward.

- Keep the myopic worker problem and effort displacement.
- Allow assistance to increase learning from remaining effort.
- Derive exact two-state conditions and test simulated dynamics.

The contribution is a stationary-distribution characterization. It does not establish a welfare or productivity ranking.

Baseline: arXiv:2605.11350v1, Prop. 3.5, p. 8. Learning complementarity itself is not a novelty claim; see the topic proposal.

The worker's decision is unchanged

For fixed skill s and assistance $a \geq 0$,

$$\max_{e \geq 0} \; p(s + e + a) - \gamma e, \qquad \gamma > 0.$$

p is continuously differentiable, increasing and concave; $\limsup_{x \to \infty} p(x)/x < \gamma$ ensures finite maximizers.

$$x^* = \max \arg\max_{x \geq 0} \{p(x) - \gamma x\}, \qquad e^*(s, a) = (x^* - s - a)^+.$$

Retain the original **largest-maximizer selection** if choices tie.

$$e > 0 : \; p'(s + e + a) = \gamma; \qquad e = 0 : \; p'(s + a) \leq \gamma.$$

Learning does not enter this objective. Its technology cannot change the FOC.

$$0 \le s_L < s_H, \qquad b = x^* - s_L > 0, \qquad e_L^*(a) = (b-a)^+.$$

At fixed assistance, the continuous-time Markov chain has

$$Q(a) = \begin{pmatrix} -\lambda_L(a) & \lambda_L(a) \\ \mu & -\mu \end{pmatrix}, \qquad \lambda_L(a) = \lambda_0 + \kappa(b-a)^+(1+\eta a).$$

Assume $\lambda_0, \kappa, \mu > 0$ and $\eta \ge 0$ throughout. Balance and normalization give the unique stationary law:

$$(1-\pi_H)\lambda_L = \pi_H \mu \quad \Longrightarrow \quad \boxed{\pi_H(a) = \frac{\lambda_L(a)}{\lambda_L(a)+\mu}.}$$

Positive rates imply irreducibility. Effort in H does not affect transitions because there is no higher state.

With positive effort, $0 \leq a < b$,

$$\lambda_L(a) = \lambda_0 + \kappa(b-a)(1+\eta a).$$

The marginal effect splits into two terms:

$$\lambda_L'(a) = \kappa \left[\underbrace{\eta(b-a)}_{\text{learning from remaining effort}} - \underbrace{(1+\eta a)}_{\text{lost human effort}} \right] = \kappa[\eta(b-2a)-1].$$

Complementarity must exceed displacement, rather than merely be positive. At $a \geq b$, effort is zero and $\lambda_L(a) = \lambda_0$.

Proposition 1: global stationary skill

Under the worker and rate assumptions just stated:

- If $\eta b \leq 1$, π_H strictly decreases on $[0, b]$, then is constant. Its unique maximum is at $a = 0$.
- If $\eta b > 1$, define

$$a^* = \frac{b - 1/\eta}{2}, \qquad a^R = b - \frac{1}{\eta} = 2a^*.$$

  Then $0 < a^* < a^R < b$: π_H strictly increases on $[0, a^*]$, strictly decreases on $[a^*, b]$, and is constant on $[b, \infty)$.
- The unique global maximum is a^* in the strong regime.

At $\eta b = 1$, the right derivative at zero is zero, but there is no increasing interval.

Three different comparisons

In the strong regime $\eta b > 1$:

Question	Exact answer
Positive marginal effect	$0 \leq a < a^*$ (right derivative at zero)
Better than no AI	$0 < a < a^R$
Better than lower positive AI	depends on both levels
Zero human effort	$a \geq b$

For $a^* < a < a^R$, assistance is already reducing stationary skill, while skill remains above the no-AI benchmark.

$$\pi_H(a^R) = \pi_H(0), \qquad \pi_H(a \geq b) = \frac{\lambda_0}{\lambda_0 + \mu} < \pi_H(0).$$

Proof: signs, maximum, and the effort boundary

On the active-effort region,

$$\pi'_H(a) = \frac{\mu\kappa[\eta(b-2a)-1]}{[\lambda_L(a)+\mu]^2}.$$

Positivity of μ, κ and the denominator transfers the sign to $\eta(b-2a)-1$, which strictly decreases when $\eta > 0$.

- Its root is interior exactly when $\eta b > 1$.
- Monotonicity plus the constant tail establishes the global maximum.
- At b, continuity holds, but differentiability fails:

$$\pi'_H(b-) = -\frac{\mu\kappa(1+\eta b)}{(\lambda_0+\mu)^2} < 0, \qquad \pi'_H(b+) = 0.$$

The active branch is strictly concave; the full curve is not globally concave.

Proposition 2: comparing two assistance levels

For $0 \leq a_\ell < a_h < b$, under the same assumptions,

$$\pi_H(a_h) - \pi_H(a_\ell) = \frac{\mu\kappa(a_h - a_\ell)[\eta(b - a_h - a_\ell) - 1]}{[\lambda_L(a_h) + \mu][\lambda_L(a_\ell) + \mu]}.$$

Therefore

$$\boxed{\pi_H(a_h) > \pi_H(a_\ell) \iff \eta(b - a_h - a_\ell) > 1.}$$

In the strong regime this is $a_\ell + a_h < a^R$. Set $a_\ell = 0$ to obtain the no-AI return threshold.

If $a_\ell < b \leq a_h$, skill strictly falls; if $b \leq a_\ell < a_h$, the shares are equal. In two states, share ordering is equivalent to FOSD.

Numerical design: three illustrative regimes

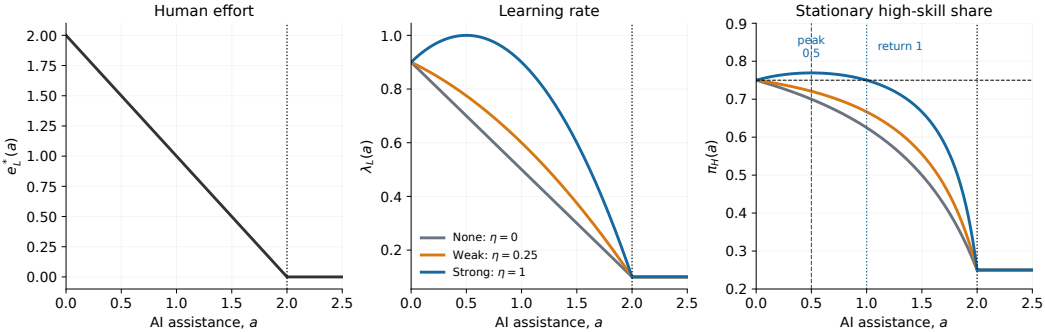
Use $p(x) = 4\log(1+x)$, $\gamma = 1$, $s_L = 1$, $s_H = 3$:

$$x^* = 3, \quad b = 2, \qquad \lambda_0 = 0.1, \quad \kappa = 0.4, \quad \mu = 0.3.$$

Regime	η	ηb	Predicted shape
No complementarity	0	0	decreasing
Weak complementarity	0.25	0.5	decreasing
Strong complementarity	1	2	increasing, then decreasing

Rates are per common time unit; a, s, e share an input unit and η has its inverse unit.
Parameters expose different regimes; they are not estimated from data.

Numerical comparative statics



Effort falls identically in every regime. Only strong complementarity initially raises learning and stationary skill.

Closed-form evaluations from code/verify.py; these lines are not stochastic simulations.

The strong regime: a small, bounded skill gain

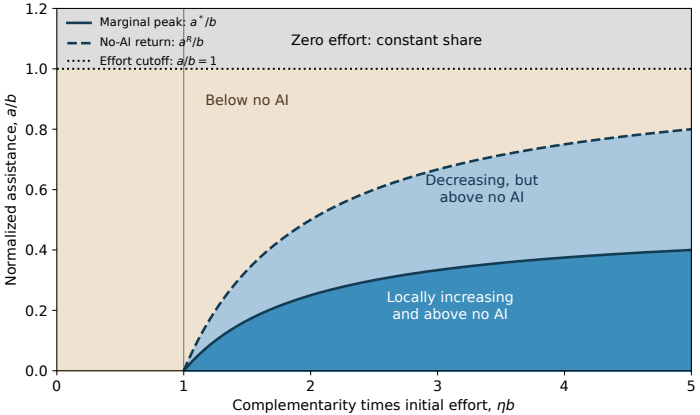
Assistance	Effort	Learning rate	Stationary high skill
0	2	0.9	75.00%
$a^* = 0.5$	1.5	1.0	76.92%
$a^R = 1$	1	0.9	75.00%
$b = 2$	0	0.1	25.00%

The peak gain is 1.9231 percentage points with 25% less effort than without AI. This is a skill result, not an optimal assistance policy.

$$\lambda_L(a^*) - \lambda_L(0) = \frac{\kappa(\eta b - 1)^2}{4\eta}.$$

Near $\eta b = 1$, the gain is second-order small.

Threshold sensitivity: complementarity and baseline effort



Closed-form regime map. The return boundary differs from the marginal-effect boundary.

Let $z = a/b$ and $\theta = \eta b$.

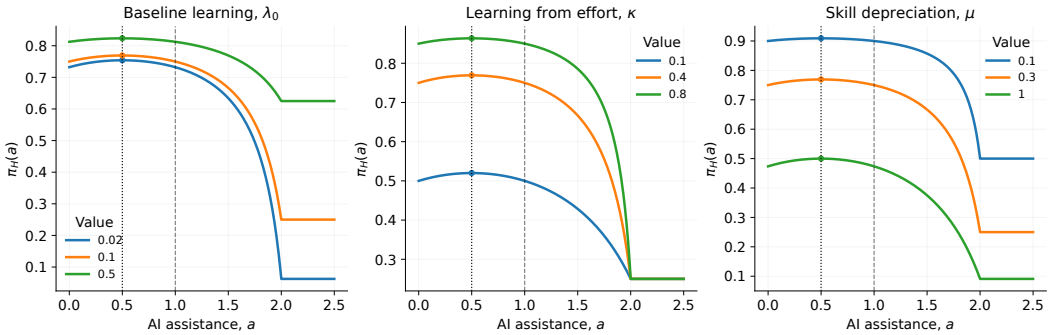
$$z^* = \tfrac{1}{2}(1 - 1/\theta)$$

$$z^R = 1 - 1/\theta$$

These exist positively only for $\theta > 1$.

Higher baseline effort b lowers the critical complementarity $1/b$.

Rates affect magnitudes, not threshold locations



Thresholds depend only on (b, η). Rates change levels and gains. A common scaling of (λ_0, κ, μ) leaves $\pi_H(a)$ unchanged, but speeds convergence. Gain size need not rise with κ or μ.

Parameter sweeps and rate-scaling invariance checked numerically; sensitivity results proved in the manuscript.

Stochastic experiment: what is actually simulated

For each fixed a, simulate the CTMC with exponential holding times of means $1/\lambda_L$ in L and $1/\mu$ in H.

$$\widehat{A}_T = \frac{1}{T}\int_0^T 1_{\{S_t=H\}}\, dt.$$

- 384 independent paths per cell; seed 20261104.
- Start in L; horizons $T = 25, 100, 400$, with no burn-in.
- Include the last partial holding interval; no time discretization.
- Estimate standard errors across independent path occupancies.

A count of visits would mislead: the two-state jump chain alternates. We measure the time spent in each state.

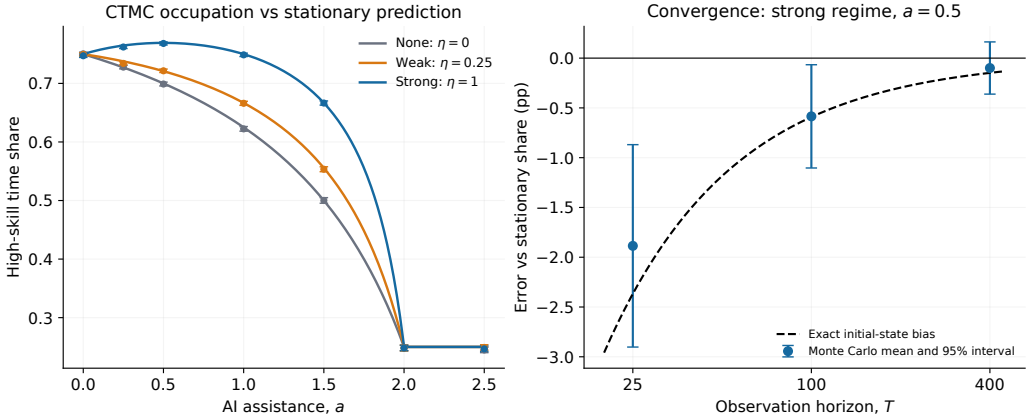
Write $r = \lambda_L + \mu$ and start in L. Exact predictions are

$$\Pr(S_t = H) = \pi_H(1 - e^{-rt}), \qquad E[\widehat{A}_T] = \pi_H \left[1 - \frac{1 - e^{-rT}}{rT}\right].$$

$$\underbrace{\overline{A}_T - \pi_H}_{\text{gap from stationarity}} = \underbrace{\overline{A}_T - E[\widehat{A}_T]}_{\text{Monte Carlo error}} + \underbrace{E[\widehat{A}_T] - \pi_H}_{\text{initial-state bias}}.$$

We check means against the finite-horizon prediction using exact occupation variance, and report nominal 95% intervals from empirical SEs. The automated tolerance is six exact SEs; it is a diagnostic, not proof.

CTMC occupation approaches the analytic prediction



At $T = 400$, the largest gap from stationary skill is 0.5223 pp; the largest standardized error against the finite-horizon mean is 1.8486.

Independent stochastic paths; nominal 95% Monte Carlo intervals do not measure parameter uncertainty.

What the computational evidence establishes

- Symbolic factorization checks derivatives, comparisons, and variance.
- Independent effort optimization checks the unchanged worker choice.
- Numerical checks cover equality, effort boundaries, arbitrary pairs, parameter sensitivity, and stationary balance.
- CTMC experiments test occupation against exact dynamics; increasing T reduces initial-state bias and sampling error in scale.

These are illustrative model checks. They do not estimate complementarity, prove the theorem, or identify effects in data.

Single entry point: python code/verify.py. Outputs, seeds, tolerances, and environment are recorded in code/output/.

Completed mathematically: effort selection, unique stationarity, transient probabilities, global shape, exact two-level comparisons, thresholds, and rate-scaling invariance.

Delicate steps: preserve tie-breaking under weak concavity; handle $\eta = 0$, $\eta b = 1$, and the kink at b; distinguish a share level from a gain.

Open research scope: endogenous assistance, forward-looking effort, empirical identification, and more than two skill states. No result for a general N-state chain is claimed.

Status: the official formalization has not been run.

Mathematical object/result	Required formal counterpart
Selected optimum and $(b-a)^+$	optimizer with original tie rule
Balance and stationary law	generator and unique invariant law
Proposition 1	sign regions, maximum, return, boundary
Proposition 2	exact comparison with positive denominators

Every numbered result needs an explicit status. No conclusion may be introduced as a hypothesis.

Lean: evidence pending the official workflow

No verified Lean declaration or proof fragment is available today.

The sequence required by Issue #7 is:

1. Review and merge the chosen paper version; pin its commit.
2. Run AppliedModelingLib's paper-formalization skill.
3. Retain the scoped check and copy the entire generated folder.
4. Show the actual statement, proof fragment, and exact scope here.

The Lean requirement remains incomplete. Unexecuted code would not certify our result.

Workflow: lean/README.md. Merge awaits the author's review.

AI-assisted result to check: for two active-effort levels,

$$\Delta\pi_H \text{ has sign } \eta(b - a_h - a_\ell) - 1.$$

Let $g(a) = (b - a)^+(1 + \eta a)$. With $b = 2$, $\eta = 1$, $a_\ell = 0.4$, $a_h = 0.8$:

$$g(a_\ell) = 2.24 > g(a_h) = 2.16 > g(0) = 2.$$

Algebraic/computational verdict: the pairwise condition holds. The example shows why $\eta b > 1$ alone cannot rank every pair.

Author's handwritten derivation and verdict: pending. This typed calculation is not handwritten evidence.

Author must independently derive the fraction subtraction and sign condition. Checklist: hand/README.md.

Questions for class critique

1. Is learning complementarity a convincing mechanism when the worker remains myopic, or should the next model make effort forward-looking?
2. Which observable learning outcome could distinguish η from background learning and effort effectiveness?
3. Is the bounded increase in stationary skill economically meaningful, and which additional outcome should accompany it before discussing policy?